

CORE: Uncertainty-Gated Policy Optimization for Robust Contact-Rich Manipulation

Abstract

We present CORE, an uncertainty-gated policy optimization method for contact-rich manipulation under distribution shift.

In the current multiseed benchmark slice, our method reaches 0.7494 success versus baseline 0.7119 (+0.0375).

Results are currently benchmark-scoped and include explicit failure and limitation tracking.

1. Introduction

Robust contact-rich manipulation under distribution shift remains difficult to evaluate and iterate safely.

2. Related Work

Recent robust planning/manipulation work is tracked in weekly literature briefs and mapped to novelty comparisons.

3. Method

CORE combines uncertainty-penalized trust-region updates with an online promote/rollback gate to reject unstable policy steps.

4. Experimental Setup

We use fixed-seed suites and consistent run budgets across baseline and method variants.

5. Results

Main benchmark: baseline=0.7119, method=0.7494, delta=+0.0375.

Ablation and robustness results are reported in WS5 artifacts.

6. Discussion and Limitations

Current evidence is strong for the benchmark stack with software-transfer stress validation, while exact upstream baseline parity remains future work.

7. Conclusion

Uncertainty-gated updates improve robustness in benchmark-scoped software validation and support conservative, evidence-backed claims.